

Exponents and Surds — Complete Notes

Grade 11

Mathematics

CAPS

What you'll learn

- Apply the laws of exponents to simplify expressions with integer and rational exponents.
- Switch between exponent form and surd (radical) form.
- Simplify, add and multiply surds, and rationalise the denominator of a fraction.
- Solve exponential equations by matching bases or by using a substitution.
- Solve surd equations and recognise answers that must be rejected.

Exponents: the language of growth

An exponent tells you how many times a base is multiplied by itself. In a^n , the number a is the base and n is the exponent (also called the index). This short notation hides a lot of power: it lets us write very large and very small numbers, describe growth and decay, and solve equations that would be impossible by hand. The whole topic rests on a handful of laws, and every one of them can be proved by simply writing the powers out in full.

In the South African CAPS curriculum, this work sits in Paper 1 and turns up again later inside financial mathematics, functions and logarithms. Getting the laws automatic now saves you time in every later topic.

The laws of exponents

Law	Rule	Worked illustration
Product	$a^m \times a^n = a^{(m+n)}$	$x^3 \times x^4 = x^7$
Quotient	$a^m \div a^n = a^{(m-n)}$	$x^5 \div x^2 = x^3$
Power of a power	$(a^m)^n = a^{(mn)}$	$(x^2)^3 = x^6$
Power of a product	$(ab)^n = a^n \times b^n$	$(2x)^3 = 8x^3$
Zero exponent	$a^0 = 1$ ($a \neq 0$)	$7^0 = 1$
Negative exponent	$a^{(-n)} = 1 \div a^n$	$2^{-3} = 1/8$
Rational exponent	$a^{(1/n)} = \sqrt[n]{a}$	$8^{(1/3)} = \sqrt[3]{8} = 2$
Fractional exponent	$a^{(m/n)} = \sqrt[n]{(a^m)} = (\sqrt[n]{a})^m$	$27^{(2/3)} = 9$

The rule of thumb: whenever the bases are the same and the operation is multiplication or division, the exponents add or subtract. Everything else follows from that one idea.

Negative and rational exponents

A negative exponent does not make the answer negative. It flips the base to the other side of the fraction bar, so $5^{-2} = 1/25$, not -25 .

A fractional exponent connects exponents to roots: the denominator of the fraction is the root you take, and the numerator is the power you raise it to. For example, $32^{(2/5)} = (\sqrt[5]{32})^2 = 2^2 = 4$. This is why the laws of exponents and the laws of surds are really the same laws wearing different clothes.

Surds: irrational roots

A surd is a root that cannot be simplified to a rational number. $\sqrt{2}$, $\sqrt{7}$ and $\sqrt[3]{5}$ are surds because their decimal expansions go on forever without repeating. $\sqrt{9}$ is not a surd, because it equals exactly 3.

The key surd rules mirror the exponent laws:

- $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$
- $\sqrt{a \div b} = \sqrt{a} \div \sqrt{b}$, for $b > 0$
- $(\sqrt{a})^2 = a$

Only like surds can be added or subtracted directly. $\sqrt{2} + \sqrt{3}$ cannot be simplified any further, but $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$. To combine surds that look different, break each one down and pull out the biggest perfect-square factor: $\sqrt{50} = \sqrt{(25 \times 2)} = 5\sqrt{2}$.

Rationalising the denominator

Textbook and exam answers avoid surds sitting in the denominator of a fraction. To remove a single surd, multiply the top and bottom by that same surd. To remove a two-term denominator such as $(\sqrt{a} - \sqrt{b})$, multiply top and bottom by its conjugate $(\sqrt{a} + \sqrt{b})$; the difference of squares then clears both roots in one step.

$$\begin{array}{lclcl} \frac{a}{\sqrt{b}} & = & \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} & = & \frac{a\sqrt{b}}{b} \quad \text{(single surd below)} \\ \\ \frac{1}{\sqrt{a} - \sqrt{b}} & = & \frac{1}{\sqrt{a} - \sqrt{b}} \times \frac{(\sqrt{a} + \sqrt{b})}{(\sqrt{a} + \sqrt{b})} & = & \frac{\sqrt{a} + \sqrt{b}}{a - b} \quad \text{(conjugate)} \end{array}$$

Worked example 1: Simplifying with exponent laws

Simplify $(2x^3y^{-2})^3 \times 3x^2y^4 \div 6x^5y^{-1}$.

Step 1 — Remove the bracket using the power-of-a-product and power-of-a-power laws. $(2x^3y^{-2})^3 = 2^3 \times x^{(3 \times 3)} \times y^{(-2 \times 3)} = 8x^9y^{-6}$.

Step 2 — Multiply by the next factor. Multiply the coefficients and add the exponents of like bases. $8x^9y^{-6} \times 3x^2y^4 = 24 \times x^{(9+2)} \times y^{(-6+4)} = 24x^{11}y^{-2}$.

Step 3 — Divide by $6x^5y^{-1}$. Divide the coefficients and subtract the exponents. $24 \div 6 = 4$. $x^{11} \div x^5 = x^{(11-5)} = x^6$. $y^{-2} \div y^{-1} = y^{(-2-(-1))} = y^{-1} = 1/y$.

Answer: $4x^6 \div y$, which we usually write as $4x^6/y$.

Worked example 2: An exponential equation that needs a substitution

Solve $3^{2x} - 10 \times 3^x + 9 = 0$.

Step 1 — Notice that 3^{2x} equals $(3^x)^2$. The equation is quadratic in disguise. Let $t = 3^x$, where $t > 0$ because a positive base raised to any power is always positive.

Step 2 — Substitute to get a quadratic in t . $t^2 - 10t + 9 = 0$.

Step 3 — Factorise. $(t - 1)(t - 9) = 0$, so $t = 1$ or $t = 9$.

Step 4 — Replace t with 3^x and solve for x . $3^x = 1$ gives $x = 0$. $3^x = 9 = 3^2$ gives $x = 2$.

Step 5 — Check both answers in the original equation. $x = 0$: $3^0 - 10 \times 3^0 + 9 = 1 - 10 + 9 = 0$. Correct. $x = 2$: $3^4 - 10 \times 3^2 + 9 = 81 - 90 + 9 = 0$. Correct.

Answer: $x = 0$ or $x = 2$.

Worked example 3: Simplifying and rationalising surds

Simplify $\sqrt{50} - \sqrt{18} + \sqrt{8}$, then rationalise $6 \div (\sqrt{5} - \sqrt{2})$.

Step 1 — Factor out the largest perfect square from each surd. $\sqrt{50} = \sqrt{(25 \times 2)} = 5\sqrt{2}$. $\sqrt{18} = \sqrt{(9 \times 2)} = 3\sqrt{2}$. $\sqrt{8} = \sqrt{(4 \times 2)} = 2\sqrt{2}$.

Step 2 — Collect the like surds. $5\sqrt{2} - 3\sqrt{2} + 2\sqrt{2} = (5 - 3 + 2)\sqrt{2} = 4\sqrt{2}$.

Step 3 — For the fraction, multiply the top and bottom by the conjugate $\sqrt{5} + \sqrt{2}$. $6/(\sqrt{5} - \sqrt{2}) \times (\sqrt{5} + \sqrt{2})/(\sqrt{5} + \sqrt{2})$.

Step 4 — Simplify the new denominator using difference of squares. $(\sqrt{5})^2 - (\sqrt{2})^2 = 5 - 2 = 3$.

Step 5 — Write the answer in simplest form. $6(\sqrt{5} + \sqrt{2})/3 = 2(\sqrt{5} + \sqrt{2}) = 2\sqrt{5} + 2\sqrt{2}$.

Solving surd equations

When the unknown sits under a root, isolate the root and then square both sides. Because squaring can create solutions that do not fit the original equation, you must substitute every answer back into the original equation. If the square root is set equal to a negative number, that answer is rejected immediately.

For example, $\sqrt{x + 7} = x - 5$. Squaring both sides gives $x + 7 = (x - 5)^2$, so $x + 7 = x^2 - 10x + 25$, which rearranges to $x^2 - 11x + 18 = 0$ and factorises to $(x - 2)(x - 9) = 0$. So $x = 2$ or $x = 9$. Testing $x = 2$ gives $\sqrt{9} = 3$ on the left but $2 - 5 = -3$ on the right, which is false, so we reject it. Testing $x = 9$ gives $\sqrt{16} = 4$ and $9 - 5 = 4$, which is true. The only solution is $x = 9$.

Common mistakes

- Adding exponents when you should multiply them: $a^m \times a^n = a^{(m+n)}$, not $a^{(mn)}$.
- Thinking $a^{(m+n)}$ equals $a^m + a^n$. It does not — $2^{3+2} = 32$, while $2^3 + 2^2 = 12$.
- Writing $(a + b)^2 = a^2 + b^2$. The correct expansion is $a^2 + 2ab + b^2$.
- Making a negative exponent negative: $3^{-2} = 1/9$, not -9 .
- Believing $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$. Test it: $\sqrt{9 + 16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.
- Skipping the check step in surd equations, which lets false or extraneous roots survive.
- Forgetting that $a^0 = 1$ for every non-zero base, and that 0^0 is undefined.

Quick summary / cheat sheet

Idea	Remember this
Same base, multiply	add the exponents
Same base, divide	subtract the exponents
Bracket raised to a power	multiply the exponents
Negative exponent	write it as a reciprocal
Fractional exponent	denominator is the root, numerator is the power
Simplify a surd	pull out the biggest perfect square
Adding surds	only like surds can combine
One-surds denominator	multiply top and bottom by that surd
Two-term denominator	multiply top and bottom by the conjugate
Surd equation	square both sides, then test every answer

Also keep these two relationships close:

- $\sqrt{ab} = \sqrt{a} \sqrt{b}$ and $\sqrt{a \div b} = \sqrt{a} \div \sqrt{b}$.
- To solve an exponential equation, either match the bases so the exponents are equal, or substitute t for the repeated power.

Practice questions

1. Simplify $(3x^4y^{-3})^2 \div 9x^5y^{-4}$.
2. Simplify $16^{(3/4)}$.
3. Solve $5^{(2x-1)} = 125$.
4. Solve $2^{(2x)} - 5 \times 2^x + 4 = 0$.
5. Simplify $\sqrt{75} + \sqrt{12} - \sqrt{27}$.
6. Rationalise $8 \div (\sqrt{7} + \sqrt{3})$.
7. Solve $\sqrt{(2x + 1)} = x - 1$.
8. Simplify $2^{-3} \times 2^5 \div 2^0$.

Answers

1. $(3x^4y^{-3})^2 = 9x^8y^{-6}$. Divide by $9x^5y^{-4}$: the coefficient becomes $9 \div 9 = 1$; $x^{(8-5)} = x^3$; $y^{(-6-(-4))} = y^{-2}$.
Answer: x^3/y^2 .
2. $16^{(3/4)} = ({}^4\sqrt{16})^3 = 2^3 = 8$.
3. $125 = 5^3$, so $2x - 1 = 3$, giving $2x = 4$ and $x = 2$.
4. Let $t = 2^x$: $t^2 - 5t + 4 = 0$, so $(t - 1)(t - 4) = 0$, giving $t = 1$ or $t = 4$. Then $2^x = 1$ gives $x = 0$, and $2^x = 4$ gives $x = 2$. Answer: $x = 0$ or $x = 2$.
5. $\sqrt{75} = 5\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$. So $5\sqrt{3} + 2\sqrt{3} - 3\sqrt{3} = 4\sqrt{3}$.
6. Multiply top and bottom by $(\sqrt{7} - \sqrt{3})$: the denominator becomes $7 - 3 = 4$, so the answer is $8(\sqrt{7} - \sqrt{3})/4 = 2(\sqrt{7} - \sqrt{3}) = 2\sqrt{7} - 2\sqrt{3}$.
7. Squaring gives $2x + 1 = (x - 1)^2 = x^2 - 2x + 1$, so $x^2 - 4x = 0$ and $x(x - 4) = 0$, giving $x = 0$ or $x = 4$. Test $x = 0$: the left side is $\sqrt{1} = 1$ but the right side is $0 - 1 = -1$, so reject it. Test $x = 4$: $\sqrt{9} = 3$ and $4 - 1 = 3$, so

accept it. Answer: $x = 4$.

8. $2^{(-3+5-0)} = 2^2 = 4$.

Official EduRank Team notes. Always cross-check with your textbook and teacher.